

Derivative Pricing

(3. Change of Measure)



Continuous version – Sneak Peak



From Discrete to Continuous (I)

Binomial model for stock S in limit approaches the continuous model for S . We have already seen the convergence of scaled symmetric random walk to Brownian motion. Let's consider an n -period binomial tree with each time step $\delta t = t/n$. We define the up-move and down-move factors as

$$\begin{matrix} \nearrow (\sqrt{\delta t}) \\ \searrow -\sqrt{\delta t} \end{matrix}$$

$$\begin{aligned} &\rightarrow \begin{cases} \exp(\mu \delta t + \sigma \sqrt{\delta t}) & \text{if up} & p = 1/2 \\ \exp(\mu \delta t - \sigma \sqrt{\delta t}) & \text{if down} & 1-p = 1/2 \end{cases} \end{aligned}$$

Assume under real-world probability measure $\mathbb{P}$, the up-move probability is $p = 1/2$ in every step. After n steps, let's say there were total X_n number of up moves, $(n - X_n)$ down moves. The stock price at t is

$$(\delta t)$$

$$(S_t) = S_0 \exp \left[\mu t + \sigma \sqrt{t} \left(\frac{2X_n - n}{\sqrt{n}} \right) \right]$$

$$n \rightarrow \infty, \delta t = t/n \rightarrow 0$$

Prove the above result and show that S_t asymptotically approaches $S_t = S_0 \exp[\mu t + \sigma Z \sqrt{t}]$

From Discrete to Continuous (II)

What is the distribution of S_t under martingale measure $\mathbb{Q}$? We know that the probability of up move under martingale measure is

$$\underline{q} = \frac{e^{r\delta t} - d}{u - d}$$

In our model, for $\delta t \rightarrow 0$, q evaluates to

$$\underline{q} = \frac{1}{2} \left[1 - \sqrt{\delta t} \left(\frac{\mu - r + \frac{1}{2}\sigma^2}{\sigma} \right) \right]$$

Under this measure the stock price has the following terminal distribution with $\mathbb{E}^{\mathbb{Q}}(S_t) = S_0 e^{rt}$ which we know so well by now. (Real world drift is gone)

$$e^{-rT} (S_T - K)^+$$

$$S_t = S_0 \exp \left[\left(r - \frac{1}{2}\sigma^2 \right) t + \sigma \sqrt{t} Z \right]$$

The claim of a call option $X = (S_T - K)^+$ 

Carry out the integral to evaluate the expectation above if you are feeling adventurous! The answer to this is the famous BSM formula

Φ is the cumulative normal distribution.



Change of Measure



Change of Measure (CMG)(I)

We have made a clear distinction between **process** and **measure**. Ito calculus helps us manipulate stochastic processes and work on differentials of processes, but it does not give us the algebra for manipulating the measure. Manipulating the measure is a key requirement as we have already seen that for pricing, we need to go from $\mathbb{P}$ to $\mathbb{Q}$. This topic is dedicated to developing necessary infrastructure to change the measure.

$N(0, t)$, $N(0, dt)$

dY dW

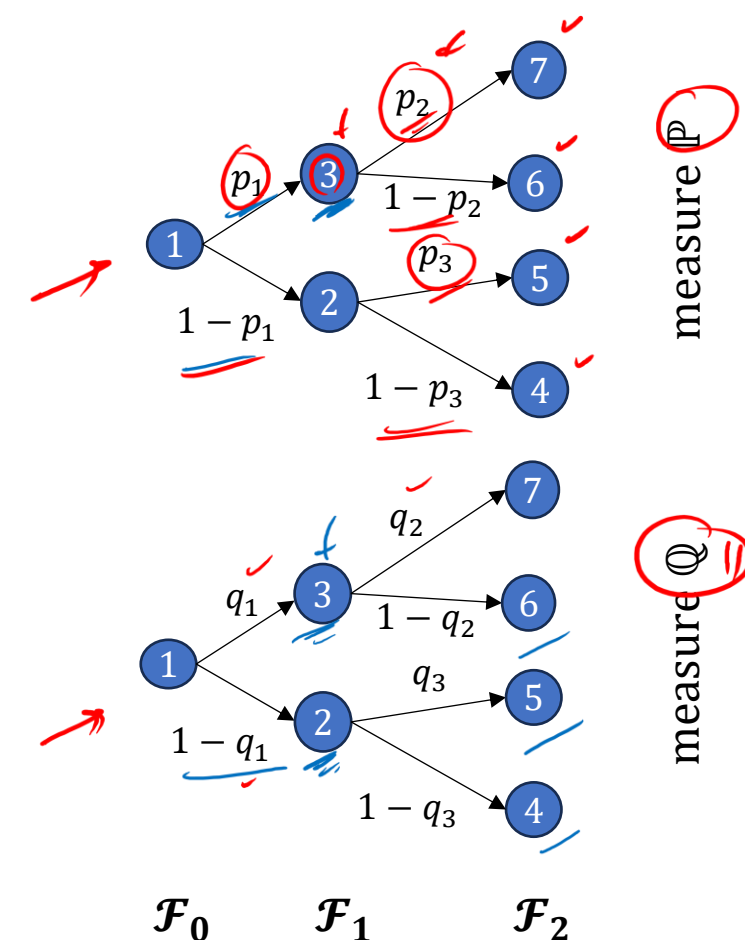
When we say W_t is Brownian, question is under which measure ? We must bring the measure into the picture and say W_t is Brownian under say measure $\mathbb{P}$, or W_t is a $\mathbb{P}$ -Brownian.

When we change the measure say from $\mathbb{P}$ to $\mathbb{Q}$, W_t may not be Brownian.

To understand change of measure, we can refer to our binomial setting.

Change of Measure (CMG)(II)

Consider the following 2-step binomial tree under two different probability measures $\mathbb{P}$ and $\mathbb{Q}$. Let's denote the path probabilities under the two measures as follows



π path prob. $\rightarrow$ π' path prob.

Node	path	under $\mathbb{P}$	under $\mathbb{Q}$
7	$\{1,3,7\}$	$(\pi_1) = p_1 p_2$ ✓	$\pi_1' = q_1 q_2$ ✓
6	$\{1,3,6\}$ -	$\pi_2 = p_1 (1 - p_2)$ ✓	$\pi_2' = q_1 (1 - q_2)$ ✓
5	$\{1,2,5\}$	$\pi_3 = (1 - p_1) p_3$ ✓	$\pi_3' = (1 - q_1) q_3$ ✓
4	$\{1,2,4\}$	$\pi_4 = (1 - p_1) (1 - p_3)$ ✓	$\pi_4' = (1 - q_1) (1 - q_3)$ ✓

Radon-Nikodym Derivative (I)

Think of going from $\mathbb{P}$ to $\mathbb{Q}$ as a distortion to the distribution. There is a formal way to do this. We do it by means of **Radon-Nikodym Derivative**. The Radon-Nikodym derivative is defined as

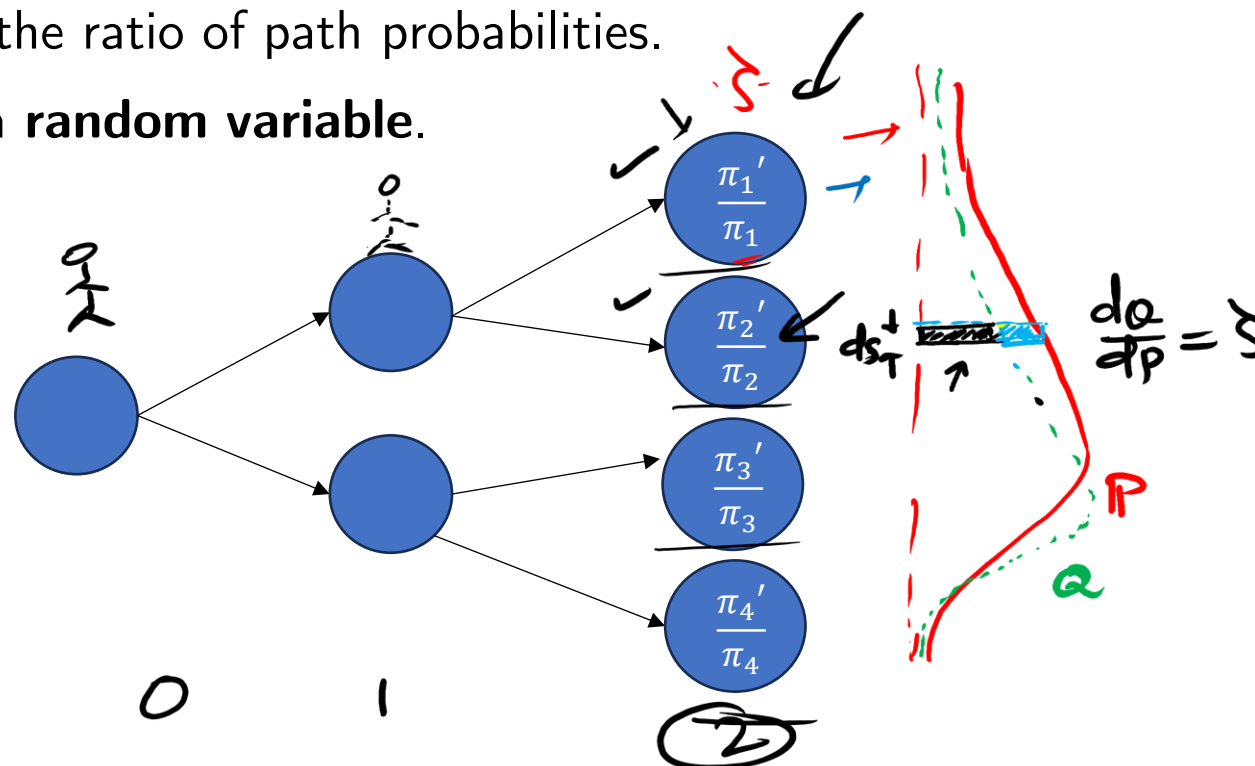
$$\zeta = \frac{d\mathbb{Q}}{d\mathbb{P}}$$

In discrete model, it's simply the ratio of path probabilities.

We can see RN derivative is a **random variable**.

$$\mathbb{P} \rightarrow \mathbb{Q}$$

$$\pi_i \times \zeta \rightarrow \pi_i'$$



$$f(x)dx = d\mathbb{P}$$

$$\int_{-a}^a d\mathbb{P} = 1$$

$$\mathbb{P}(x < X < x+dx) = d\mathbb{P}$$

Radon-Nikodym Derivative (II)

So, how to go from $\mathbb{P}$ to $\mathbb{Q}$? Given the probabilities under $\mathbb{P}$, we can calculate the probabilities under $\mathbb{Q}$ by multiplying the Radon-Nikodym Derivative. For the Radon-Nikodym derivative to be defined, the measures $\mathbb{P}$ and $\mathbb{Q}$ need to be **equivalent**.

$$\rightarrow \pi_1^{0.2} \times 5 = \pi_1^0$$

We often need to calculate expectation under $\mathbb{Q}$. How are the expectations linked ? Let X be a claim defined at time 2 which can take values $\{x_1, x_2, x_3, x_4\}$. Then,

$$\begin{aligned} \mathbb{E}^{\mathbb{P}}[X] &= \sum_i \pi_i x_i \\ \mathbb{E}^{\mathbb{Q}}[X] &= \sum_i \pi_i' x_i = \sum_i \pi_i \left(\frac{\pi_i'}{\pi_i} x_i \right) = \sum_i \pi_i \left(\frac{d\mathbb{Q}}{d\mathbb{P}} x_i \right) = \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} X \right] \end{aligned}$$

defined $\times$

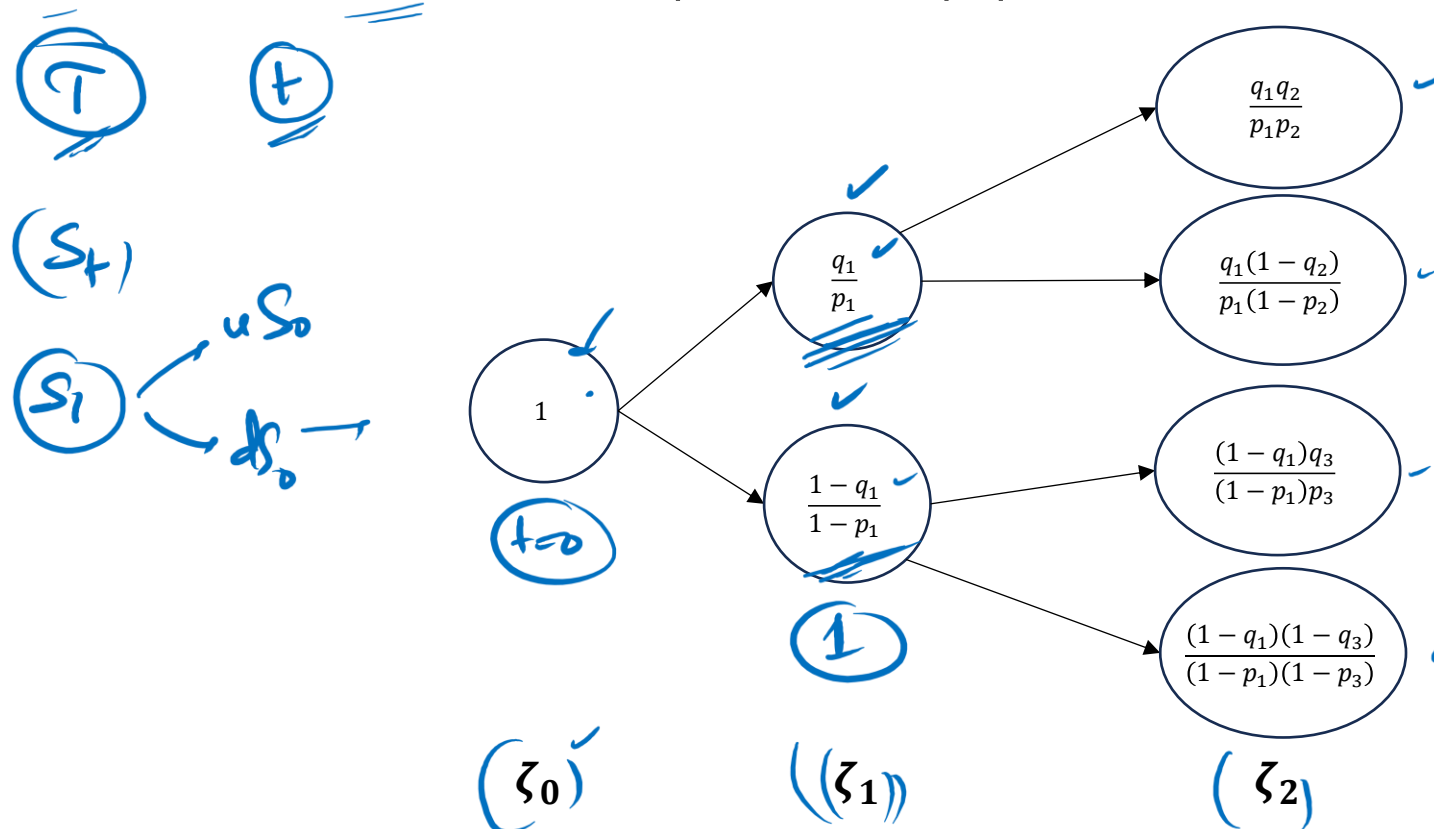
In our argument, we have defined X at T , and we are taking the unconditional expectation of X . So more formally, we can write

$$\mathbb{E}^{\mathbb{Q}}[X_T | \mathcal{F}_0] = \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} X_T | \mathcal{F}_0 \right]$$

Radon-Nikodym Derivative (III)

How about $\mathbb{E}^{\mathbb{Q}}[X_t | \mathcal{F}_s]$? We need to know $\frac{d\mathbb{Q}}{d\mathbb{P}}$ not only at the end of the paths, but everywhere.

We can make $\frac{d\mathbb{Q}}{d\mathbb{P}}$ a **process** by keeping a floating time horizon, i.e. (ζ_t) . This is the ratio of path probabilities until time t . For example, we can populate the values of ζ_t along the binomial tree.



Alternate Definition of RN derivative

There is another expression for ζ_t , it's the conditional expectation of T-horizon Radon Nikodym Derivative under $\mathbb{P}$ given $\mathcal{F}_t$. From the expression, it's easy to see that ζ_t is a $\mathbb{P}$ -martingale.

$$\rightarrow \zeta_t = \mathbb{E}^{\mathbb{P}} \left(\frac{d\mathbb{Q}}{d\mathbb{P}} \middle| \mathcal{F}_t \right) \quad (\text{verify this on the tree})$$

This ζ_t is exactly what we need. This gives us the change of measure needed for any time t , to convert the expectation from one measure to another. For example,

$$\rightarrow \mathbb{E}^{\mathbb{Q}}[X_t | \mathcal{F}_0] = \mathbb{E}^{\mathbb{P}}[\zeta_t X_t | \mathcal{F}_0]$$

And $\mathbb{E}^{\mathbb{Q}}[X_t | \mathcal{F}_s]$ can be seen as change of measure needed from s to t . So, we multiply the ratio ζ_t / ζ_s

$$\mathbb{E}^{\mathbb{Q}}[X_t | \mathcal{F}_s] = \zeta_s^{-1} \mathbb{E}^{\mathbb{P}} \left(\zeta_t X_t \middle| \mathcal{F}_s \right)$$